

Random Forest

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1 Introduction

Random Forest is one of the most powerful and widely used machine learning algorithms for classification and regression tasks. It belongs to the family of **ensemble learning methods**, which combine multiple models to obtain better predictive performance than a single model.

Instead of relying on one Decision Tree, a Random Forest builds a collection of Decision Trees and combines their predictions using a voting strategy. This approach significantly reduces overfitting while improving accuracy and robustness.

Random Forest is widely used in many real-world applications, including:

- Medical diagnosis
- Fraud detection
- Credit risk assessment
- Customer churn prediction
- Image classification
- Spam email detection

2 Why is it Called a Random Forest?

The name *Random Forest* comes from the idea of constructing a large collection (a forest) of randomly generated Decision Trees.

Consider the following analogy.

Suppose a patient visits a single doctor for a diagnosis. Although the doctor is experienced, there is always a possibility of making an incorrect decision.

Now imagine consulting one hundred independent doctors. If most doctors agree on the same diagnosis, the confidence in the final decision becomes much higher.

Random Forest follows the same principle:

- One Decision Tree acts as one expert.
- Many Decision Trees form a forest.
- Each tree makes its own prediction independently.
- The final prediction is obtained by majority voting.

Because many independent trees participate in the decision process, Random Forest generally produces more accurate and stable predictions than a single Decision Tree.

3 Decision Tree versus Random Forest

Table 1 summarizes the main differences between a Decision Tree and a Random Forest.

4 Main Idea

The main idea behind Random Forest is very simple.

Instead of constructing only one Decision Tree, the algorithm builds many different Decision Trees using randomly selected data and randomly selected features.

Each Decision Tree produces its own prediction.

Finally, all predictions are combined through a majority voting process to obtain the final class.

Figure 1 illustrates this concept.

5 Majority Voting

For classification problems, every Decision Tree predicts a class label.

The Random Forest counts the number of votes received by each class.

The class receiving the largest number of votes becomes the final prediction.

For example, suppose seven trees produce the following predictions:

The votes are

$$Yes = 5, \quad No = 2.$$

Therefore,

$$FinalPrediction = Yes$$

This voting strategy makes Random Forest much more reliable than relying on the prediction of a single Decision Tree.

Table 1: Comparison between Decision Tree and Random Forest

Characteristic	Decision Tree	Random Forest
Number of trees	One	Many
Overfitting	High	Low
Variance	High	Low
Interpretability	Easy	More difficult
Accuracy	Moderate	Usually higher
Training speed	Fast	Slower
Robustness	Lower	Higher

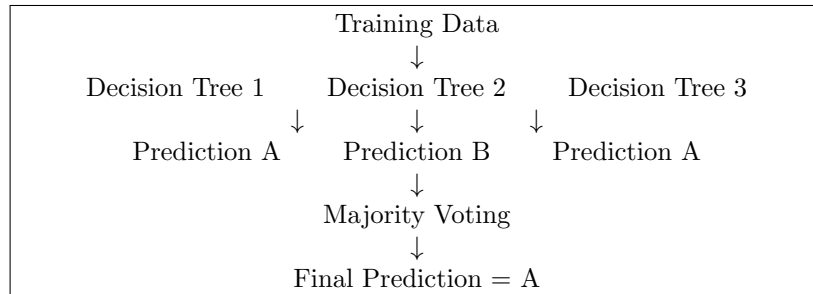


Figure 1: General idea of Random Forest.

Tree	Prediction
1	Yes
2	Yes
3	No
4	Yes
5	Yes
6	No
7	Yes

6 Mathematical Formulation

Assume that a Random Forest contains T Decision Trees.
Each tree produces a prediction

$$h_i(\mathbf{x}),$$

where

- $\mathbf{x}$ is the input feature vector,
- h_i denotes the prediction of the i -th Decision Tree.

The final prediction of the Random Forest is obtained using the mode (majority vote) of all predictions:

$$\hat{y} = \text{mode}(h_1(\mathbf{x}), h_2(\mathbf{x}), \dots, h_T(\mathbf{x})).$$

The function $\text{mode}(\cdot)$ returns the class that appears most frequently among all Decision Tree predictions.

7 Bootstrap Sampling (Bagging)

One of the key ideas behind the Random Forest algorithm is **Bootstrap Sampling**, also known as **Bagging** (Bootstrap Aggregating).

Instead of training every Decision Tree using the entire training dataset, Random Forest creates a different training dataset for each tree by randomly selecting observations from the original dataset **with replacement**.

As a result, each Decision Tree is trained on a slightly different version of the original data, making the trees less correlated and improving the overall performance of the model.

7.1 Sampling with Replacement

Bootstrap Sampling selects observations randomly from the original dataset.

After an observation is selected, it is returned to the dataset before the next selection is made. Consequently, the same observation can be selected multiple times, while some observations may not be selected at all.

This process is called **sampling with replacement**.

For example, consider the following dataset.

Suppose we generate a bootstrap sample of the same size.

Notice that

- Observation C appears twice.
- Observation B does not appear.
- The bootstrap sample still contains five observations.

This random variation is intentional and helps produce diverse Decision Trees.

7.2 Why Does Bagging Work?

If every Decision Tree were trained using exactly the same dataset, all trees would become very similar and would likely make the same prediction.

Random Forest avoids this problem by generating different bootstrap samples.

Since every tree learns from a different subset of the data, the predictions become more diverse.

Combining these diverse predictions through majority voting reduces the variance of the model and improves its ability to generalize to unseen data.

7.3 Bootstrap Sampling for Multiple Trees

Assume that the Random Forest contains three Decision Trees.

Each tree receives its own bootstrap sample.

Although every bootstrap sample contains the same number of observations, the selected observations are different.

Therefore, each Decision Tree learns a different model.

7.4 Bagging Algorithm

The bootstrap sampling procedure can be summarized as follows.

1. Begin with the original training dataset containing N observations.
2. Randomly select N observations with replacement.
3. Use the sampled observations to train one Decision Tree.
4. Repeat the process independently for every Decision Tree in the forest.

The collection of independently trained Decision Trees forms the Random Forest.

7.5 Advantages of Bagging

Bootstrap Aggregating offers several important advantages.

- Reduces overfitting.
- Reduces model variance.
- Produces more robust predictions.
- Improves generalization on unseen data.
- Increases classification accuracy when combined with multiple Decision Trees.

7.6 Important Remark

Bootstrap Sampling changes only the training data used by each Decision Tree.

Later in the Random Forest algorithm, another source of randomness is introduced by selecting only a random subset of features at each split.

These two mechanisms—bootstrap sampling and random feature selection—are what make Random Forest a powerful ensemble learning algorithm.

Table 2: Original Training Dataset

Observation
A
B
C
D
E

Table 3: Bootstrap Sample

Selected Observation
A
C
C
D
E

Table 4: Example of Bootstrap Samples

Tree	Bootstrap Sample
Tree 1	A, B, B, D, E
Tree 2	A, C, D, D, E
Tree 3	B, C, C, D, A